RESEARCH VALIDATION / CHANGE REPORT

RaspFormer

Research upgrade and complete change report

Prepared 16 October 2026. Experiments and recorded verification were completed on 15 October 2026. This document includes the implementation changes, full result tables, current figures, cleanup, reproduction commands, independent review, and scientific limits.

What changed. The project now goes beyond checking known compiler lanes. It tests geometry under rotation and compression, freezes blind write-transition predictions, applies PCA-directed causal patches, varies branch depth and branch count, and compares compiled models with 36 transformers trained from random weights.

Recorded checks	Scope
28 tests passed	Original checks plus numerical, executable-model, and recovery tests
5 notebooks / 37 code cells	Every notebook executed using the installed environment
36 checkpoints replayed	12 tasks, 3 seeds, independently checked RASP targets
7 completed reference studies	Canonical raw evidence accompanies the report

Main result. PR and CKA recover some write timing in these controlled models. That is partial transition recovery, not discovery of complete algorithms. Compression can damage the task while preserving an apparently strong transition score. Those negative cases are included.

Workspace snapshot: branch `raghulchandramouli/repo-overview`, underlying commit `2dc9118`, with local changes recorded in the manifests. Rates in this document are fractions between zero and one unless marked as percentages.

Contents

01 Repository changes 3

02 Methods and reading guide 4

03 Variance diagnosis 5

04 Blind write-transition recovery 6

05 Orthogonal rotation and scale controls 7

06 Executing residual compression 8

07 Blind recovery after transformations 9

08 Geometry-directed causal patches 10

09 Sorting row diagnosis and attention sharpening 11

10 Separate depth and branch-count controls 12

11 Learned models on the same programs 13

12 Learned versus compiled geometry 14

13 Original compiler-fidelity checks 15

14 Relation to InterpBench 16

15 Evidence and limitations 16

16 Verification and independent review 17

17 Evidence index 18

18 Decision trail 19

01 Repository changes

The twelve original program definitions are retained. Changes concentrate on numerical policy, executable model variants, research experiments, evidence recording, verification, and current documentation.

File or group	Change
raspformer/geometry.py	Adds an absolute variance floor, raw spectra and variance, and explicit active / inactive reasons for PR and CKA.
raspformer/alignment.py	Exports observed and expected variance, activity flags, numerical policy, CKA coverage, and configurable floor diagnostics.
raspformer/compiler.py	Records Git state and notebook source hashes. Uses dated runs below a runs directory and retains failed-run evidence.
raspformer/models.py	New shared JAX execution for imported compiled weights, orthogonal rotation, actual residual compression, patches, random initialization, Adam training, and NPZ checkpoints.
raspformer/research.py	New rotation / rescaling controls, compression, blind prediction and null scoring, calibration-only PCA stage selection, causal patches, sorting row errors, attention sharpening, and structural families.
raspformer/study.py	New command-line runner for the original five studies, validation controls, and learned-model comparisons. Promotes completed runs to canonical references.
raspformer/report.py	New generator for the current report and four figures. Reads preserved measurements and rejects missing local evidence links.
notebooks/05_comparison.ipynb	Uses the same comparison runner as the command line, removing duplicate numerical orchestration.
test_studies.py	Uses current canonical filenames and nested run paths while retaining original behavioral and reference checks.
test_numerics.py	New tests for near-constant signals, floor boundaries, invariance, variance exports, sampling, coverage, and provenance.
test_models.py	New tests for imported scores and residuals, rotated and compressed execution, patches, training updates, and checkpoint restoration.
test_research.py	New tests for blind detector behavior, truth isolation, sequence separation, nulls, floor sensitivity, and bounded family outputs.
scripts/check_notebooks.py	Executes notebook code cells with IPython and writes per-notebook logs and a completion summary.
scripts/replay_learning.py	Replays saved models against saved predictions and independent RASP targets. Retains compiled and learned variance in comparison tables.
scripts/verify_results.py	Checks completed manifests, frozen prediction hashes, calibration-derived stage choices, checkpoint coverage, and supplied execution logs.
README.md and .gitignore	Document commands, evidence, naming, numerical policy, interpretation, and ignored disposable runs.

Canonical evidence and cleanup

Previous artifact	Current artifact
programs/reference/step1_ground_truth_schedule.json	programs/reference/manifest.json
programs/reference/step1_ground_truth_schedule.md	programs/reference/schedule.md
trace/reference/step2_manual_trace.json	trace/reference/traces.json
Old dated report sources and duplicate bundles	results/report/research-validation.md, current figures, review, and decision trail
Absent later-stage raw tables	alignment, comparison, validation, and learning reference folders

The workspace diff marks 47 tracked files for deletion or replacement. Of those, 27 are obsolete report / archive files. The other 20 are stale canonical artifacts, including older figures and fixture names. Current evidence replaces them. The previous versions remain in Git history.

Naming uses `results/<study>/reference/` for preserved evidence and `results/<study>/runs/<UTC>-<id>/` for disposable runs. Program folders use lower-case hyphenated names such as `p12-sorting`, with learned checkpoints under `seed-42`.

02 Methods and reading guide

The core vocabulary is the integers -2 through 5. Geometry and research controls use length-six, noncausal sequences. Sampled compiler checks additionally include shorter sequences. Inputs include BOS internally; reported token observations exclude BOS.

Measurement	Meaning and limit
Participation ratio, PR	Effective covariance dimension: (sum of eigenvalues)^2 / sum of squared eigenvalues. It depends on encoding, scale, correlations, and the observation distribution.
Centered kernel alignment, CKA	Linear similarity between representations of aligned observations. Undefined when either representation fails the variance policy.
Expected representation	Independent RASP variable values placed into compiler-provided coordinates at compiler-provided write times. This supports fidelity checks, not blind discovery.
Full / computed residual views	Full includes input lanes. Computed removes named input / constant lanes but still includes compiler helper state.
Write-transition detection	Predicts that at least one variable is first written at a stage. It does not identify names, simultaneous write counts, or entire circuits.
Counterfactual match	Whether the entire patched output matches a specified symbolic variable substitution. Undefined symbolic outputs are retained as failed comparisons.

Sampling and separation

Experiment	Recorded design
Original balanced geometry	Seed 42; 64 target observations per reachable output class. Repeated targets retain weight.
Original matched comparison	3,383 shared candidate sequences at all 6 positions: 20,298 observations per program.
Compiled research controls	96 calibration sequences and 96 independent test sequences. Each set contains 576 non-BOS token observations.
Learned-model comparison	2,029 training sequences, 677 validation sequences, and 677 test sequences. Split seed 2026. Complete sequences are separated before token expansion.
Random seeds	42, 43, and 44 for learned models and rotation. Random-subspace compression uses seed 42.

The diagnostic-control splits and learned-model splits are separate designs. Positions from the same sequence are dependent. Macro means average program-level results. Seed standard deviations describe variation over the three recorded training runs, not uncertainty over all possible data distributions.

Program identifiers

ID	Task	ID	Task
P01	Absolute value	P07	Sum with next token
P02	Position parity	P08	Sum with previous token
P03	Token plus position	P09	Check nondecreasing order
P04	First element	P10	Rotate left
P05	Token histogram	P11	Check palindrome
P06	Count greater tokens	P12	Sort ascending

P01, P02, and P03 each contain one computed variable, but their balanced final computed PR values are 5, 1, and 12. Balanced one-hot support with k values has centered PR k-1. This is why PR error is retained as a secondary fidelity diagnostic rather than the main scientific outcome.

03 Variance diagnosis

The numerical policy uses total centered sample variance with ddof 1. Effective PR and component counts are zero at or below 1e-12, with an explicit inactive flag. CKA is undefined when either side is inactive. Raw variance, spectra, PR, and CKA are retained. This absolute floor is a numerical policy, not a scale-invariant definition of signal.

Matched computed-view CKA availability mismatches change from **14 to 2**. The suspicious early P07, P08, and P10 stages have variance between 8.06592e-19 and 8.06592e-19. Their raw PR is one despite negligible variation. This reproduces and resolves the normalized-score artifact at the selected floor.

Sources are matched stages [S01], floor sensitivity [S02], and balanced sensitivity [S03]. The sensitivity tables include floors from zero through 1e-8.

Program	Stage	Observed total variance	Raw PR	Filtered PR
P07	L1/Attn	8.06592e-19	1	0
P07	L1/MLP	8.06592e-19	1	0
P07	L2/Attn	8.06592e-19	1	0
P08	L1/Attn	8.06592e-19	1	0
P08	L1/MLP	8.06592e-19	1	0
P08	L2/Attn	8.06592e-19	1	0
P10	L1/Attn	8.06592e-19	1	0
P10	L1/MLP	8.06592e-19	1	0
P10	L2/Attn	8.06592e-19	1	0

Remaining mismatches are retained rather than forced to agree. They occur when observed helper state already varies before the corresponding expected named-variable state.

Program	Stage	Observed total variance	Expected total variance
P05	L1/MLP	0.732809	0.717984
P06	L1/MLP	0.91899	0.806176

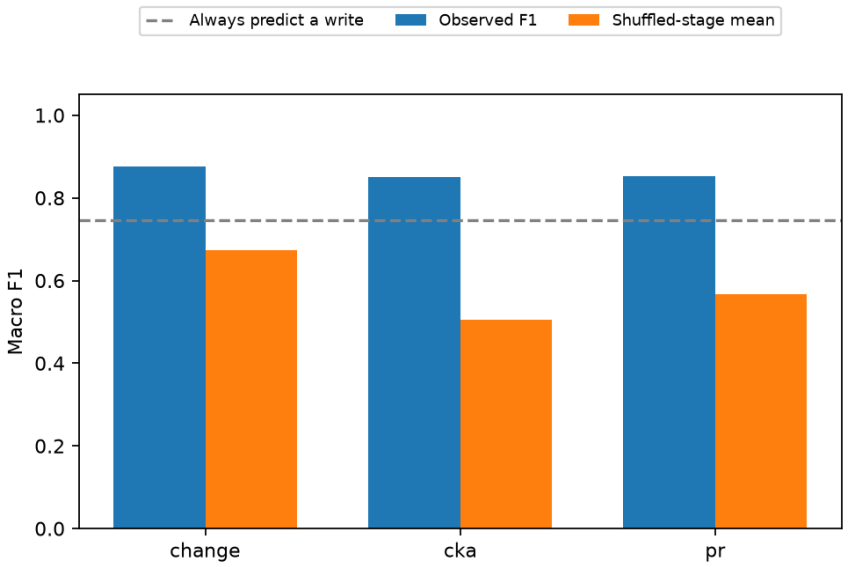
04 Blind write-transition recovery

The detector receives ordered activation matrices only. It receives no variable names, lane labels, program graph, or write schedule. Predictions are saved and hashed before schedule extraction for scoring. Thresholds are fixed. P01 through P06 form the calibration group and P07 through P12 the held-out program group.

The three detectors use residual change energy, PR change, and adjacent CKA change. The null randomly permutes each detector's predictions within a program, preserving its predicted event count, with 1,000 draws. These results concern transitions with at least one first write. They do not recover variable names, simultaneous variable counts, or complete circuits.

Group	Detector	Precision	Recall	F1	Shuffled F1	All-write F1
calibration	change	0.833333	1	0.888889	0.551556	0.666667
calibration	cka	0.916667	1	0.944444	0.511111	0.666667
calibration	pr	0.833333	1	0.888889	0.556889	0.666667
heldout	change	0.77963	1	0.875992	0.673482	0.745029
heldout	cka	1	0.748016	0.849784	0.505811	0.745029
heldout	pr	0.90119	0.837302	0.85291	0.566622	0.745029

Per-program predictions, hashes, scores, and null permutation p-values are in the validation reference [S04]. Macro means are descriptive over six programs per group. The all-positive baseline predicts a write at every transition, exposing high event prevalence.



Blind transition detection on held-out programs, with the count-matched shuffled-stage baseline.

In one held-out program-detector comparison, F1 does not exceed the always-positive baseline. These failures remain part of the result.

Program	Detector	F1	All-write F1
P11	cka	0.727273	0.736842

05 Orthogonal rotation and scale controls

These controls summarize all twelve programs. Each rotated model is also executed, so output agreement is checked separately from representation geometry.

Program	Max PR / CKA error	Executed token agreement	Import residual error
P01	6.75016e-14	1	0
P02	5.68434e-14	1	0
P03	7.10543e-14	1	0
P04	6.39488e-14	1	0
P05	8.34888e-14	1	0
P06	6.39488e-14	1	0
P07	8.17124e-14	1	0
P08	1.42109e-13	1	0
P09	7.10543e-14	1	0
P10	1.19016e-13	1	0
P11	1.22569e-13	1	0
P12	1.3145e-13	1	0

Whole-space PR and linear CKA are measured before and after square orthogonal transformations. Canonical coordinate checks deliberately retain their old lane map, exposing their dependence on basis. Nonorthogonal rescaling is a separate control. Executed rotation is measured separately because finite-precision weight transformations can amplify numerical error.

Nonorthogonal scale control

Rescaling coordinates geometrically from 0.1 to 10 changes final-stage PR on every program. Unlike orthogonal rotation, this operation changes relative feature scales. The following full-space results use the 96 diagnostic test sequences.

Program	Original PR	Scaled PR	Original / scaled CKA
P01	10.4514	4.21156	0.69723
P02	7.25541	3.62093	0.318954
P03	18.1479	5.41667	0.562194
P04	16.9444	4.63078	0.507686
P05	13.2091	4.61276	0.446789
P06	12.9308	4.61458	0.57433
P07	19.0987	7.84108	0.451831
P08	20.5534	9.63587	0.511871
P09	17.0586	6.59681	0.435633
P10	14.595	7.0484	0.422272
P11	14.5355	11.8404	0.440456
P12	26.1183	10.9098	0.74155

06 Executing residual compression

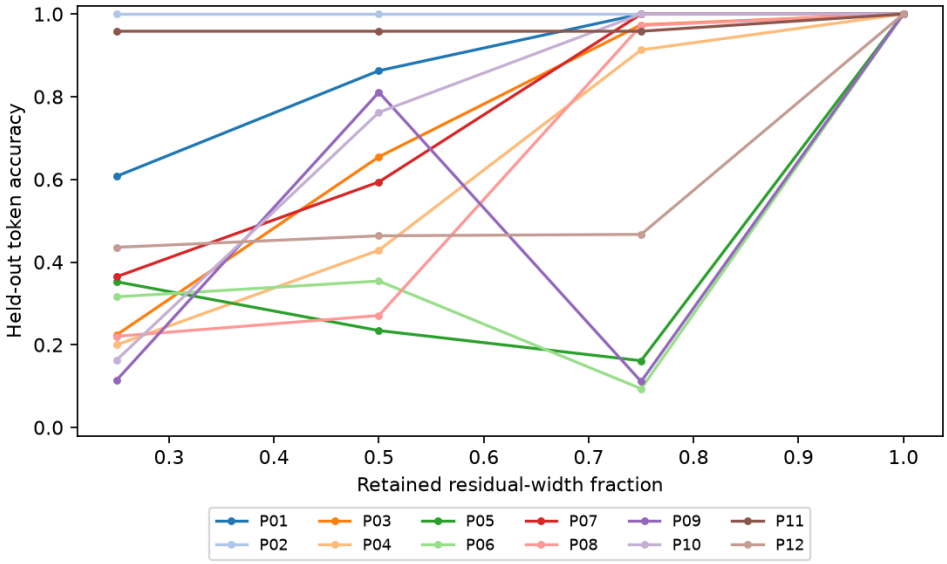
The table and chart summarize all twelve programs. The next section reports transformed recovery on the six held-out programs only.

Compression fits an uncentered shared basis on calibration sequences and transforms embeddings, residual input and output weights, biases, and unembedding. Forward execution carries the reduced residual width. Attention and MLP hidden widths are retained and reported, so residual reduction is not an equal reduction in total parameters. Random half-rank subspaces and full-rank controls are included.

The variant called pca in the stored tables uses an uncentered SVD residual basis. It includes mean structure during basis fitting; reduced ranks can still lose it. It is distinct from the centered PCA used for geometry measurements and patch directions. Requested fractions are converted to integer ranks by rounding upward.

Variant	Requested fraction	Token accuracy	Sequence accuracy	Residual MSE	Parameter fraction	Mean axis overlap
pca	0.25	0.41305	0.179688	0.0617453	0.276655	0.247447
pca	0.5	0.616175	0.277778	0.0315115	0.51773	0.121924
pca	0.75	0.72092	0.631076	0.0176273	0.766825	0.0452387
pca	1	1	1	2.24355e-12	1	1.30792e-16
random	0.5	0.370081	0.0746528	0.0967543	0.51773	0.139267

Axis overlap is the mean absolute cosine between nonzero projected original coordinate directions. Full-rank orthogonal controls have zero overlap up to rounding. Reduced models have overlapping directions, making the induced coordinate mixing measurable.



Accuracy of executing compressed models, using disjoint calibration and test sequences.

07 Blind recovery after transformations

Blind detectors are also rerun on the executing rotated and compressed states. The following table reports macro means for held-out programs P07 through P12, showing the PR and CKA detectors only. Their predictions are frozen before scoring against the original compiler schedule. When compression changes behavior, that schedule is an intended reference, not established ground truth for the changed computation. Task accuracy is therefore shown beside F1.

Variant	Requested fraction	Detector	Token accuracy	F1	Shuffled F1	All-write F1
compression-pca	0.25	cka	0.376157	0.383333	0.229233	0.745029
compression-pca	0.25	pr	0.376157	0.656227	0.384476	0.745029
compression-pca	0.5	cka	0.643229	0.598413	0.364981	0.745029
compression-pca	0.5	pr	0.643229	0.84026	0.498002	0.745029
compression-pca	0.75	cka	0.751447	0.713492	0.431251	0.745029
compression-pca	0.75	pr	0.751447	0.784704	0.460474	0.745029
compression-pca	1	cka	1	0.849784	0.505811	0.745029
compression-pca	1	pr	1	0.85291	0.566622	0.745029
compression-random	0.5	cka	0.376157	0.878884	0.554965	0.745029
compression-random	0.5	pr	0.376157	0.852259	0.637456	0.745029
rotation	1	cka	1	0.849784	0.505811	0.745029
rotation	1	pr	1	0.85291	0.566622	0.745029

08 Geometry-directed causal patches

Patch stages are selected from calibration geometric change and frozen before loading the compiler schedule. PCA fits calibration activations. A ridge decoder then maps the first three PCs to independently evaluated RASP variables. Decoder folds separate labels by whole sequence, but share an unsupervised PCA fitted on all calibration sequences. The final test sequences are independently held out from stage selection, PCA fitting, and decoder fitting. Donor-recipient pairs are sampled without selecting for different answers.

PCA ranks one and three are compared with equal-rank random subspaces, self patches, and a compiler-variable oracle. Counterfactual agreement is evaluated against a single mapped-variable substitution. Some symbolic substitutions can make RASP outputs undefined. Those cases remain explicit failures of that comparison rather than being dropped. Behavioral effects alone do not establish recovery of a named variable.

Variant	Rank	Changed output	Counterfactual match
oracle	2	0.0208333	1
oracle	6	0.8125	1
oracle	7	0.5	0.84375
oracle	8	0.8125	1
oracle	13	0.8125	1
pca	1	0.0625	0.520833
pca	3	0.135417	0.588542
random	1	0.015625	0.463542
random	3	0.0572917	0.421875
self	0	0	0.479167

The table includes 1,152 patch conditions across 192 donor-recipient pairs. The 60 rows with undefined symbolic counterfactual outputs remain failed comparisons. Self patches are a baseline-preservation check. Per-program variable decoder results and all donor identities are saved next to the patch rows.

09 Sorting row diagnosis and attention sharpening

Attention scale	Token accuracy	Max lane error	Mean lane error
1	1	0.000852227	0.000106652
2	1	7.15256e-07	8.94072e-08
4	1	1.35385e-24	1.07619e-25
8	1	0	0

At the first output write, each of the eight output lanes fails 64 of 512 probes. The failure sets are different across lanes, and their union contains all 512 probes. Equal 87.5% lane agreement therefore does not imply a shared failing eighth of observations. Attention scales multiply actual QK logits before softmax while MLP exactness stays fixed. At scale two, the maximum discrepancy is already below the lane tolerance. Decoded token accuracy remains 100%. The sweep supports a finite-attention approximation explanation. It does not establish that duplicates or ties alone cause the discrepancy.

Sources are every scalar error [S05], attention sweep [S06], and failing probe identities [S07].

10 Separate depth and branch-count controls

Branch depth	Branches	Computed variables	Realized depth	Residual width	Layers	Token accuracy
1	1	1	1	24	1	1
1	2	3	2	41	2	1
1	3	5	3	57	3	1
2	1	2	2	32	2	1
2	2	5	3	57	3	1
2	3	8	4	81	4	1
3	1	3	3	40	3	1
3	2	7	4	73	4	1
3	3	11	5	105	5	1

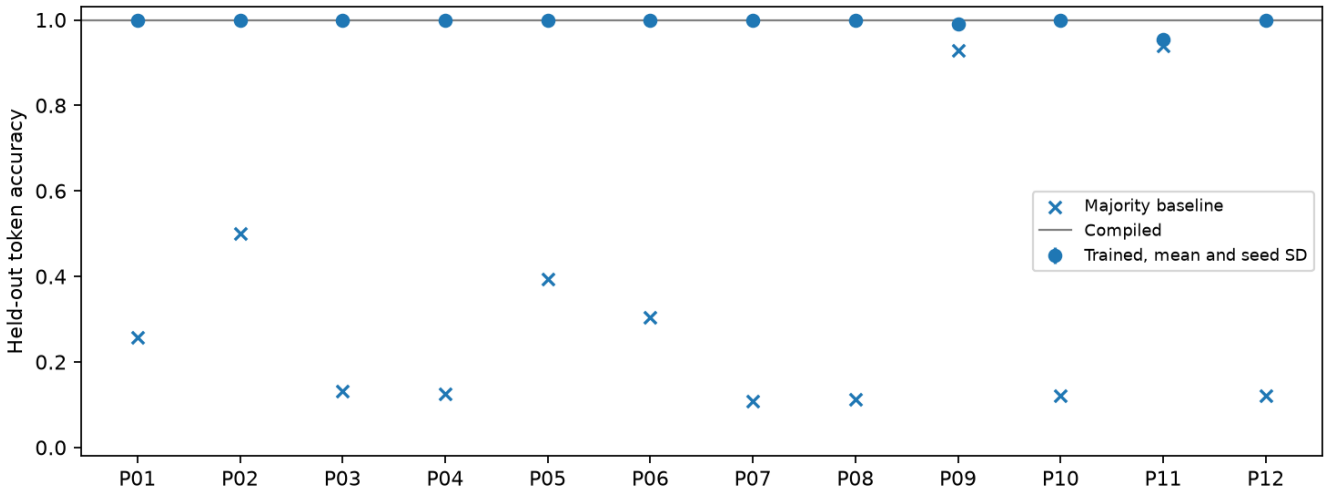
The centered realized depth and variable-count design matrix has rank **2**. Branch depth and branch count are crossed, and every branch contributes to the output. Serial merges also add depth, so requested branch depth is not identical to realized dependency depth. These token-local modulo programs are controlled structural examples, not a universal complexity scale. Their full schedules and geometry are preserved.

11 Learned models on the same programs

Program	Seeds	Token accuracy	Token SD	Sequence accuracy	Majority baseline	Compiled accuracy
P01	3	1	0	1	0.256524	1
P02	3	1	0	1	0.5	1
P03	3	1	0	1	0.130724	1
P04	3	1	0	1	0.125554	1
P05	3	0.998523	0	0.998523	0.394141	1
P06	3	0.998523	0	0.998523	0.30453	1
P07	3	0.999754	0.000246184	0.998523	0.108813	1
P08	3	0.999918	0.000142134	0.999508	0.111521	1
P09	3	0.989414	0.00390805	0.986706	0.927622	1
P10	3	1	0	1	0.121861	1
P11	3	0.954128	0.00327835	0.944362	0.939439	1
P12	3	1	0	1	0.121861	1

All 36 models start from random weights and train all parameters for 600 Adam steps. The shared architecture has residual width 32, two layers, and two attention heads. Whole unique sequences are split 60%, 20%, and 20% into train, validation, and test sets with split seed 2026. The source pool mixes random and structured length-six sequences over the original eight-token vocabulary. Results apply to this distribution and do not establish length generalization or behavior on natural language.

The majority-class baseline exposes class imbalance. Loss curves, exact predictions, sequence IDs, output encodings, and model checkpoints are saved. Learned-versus-compiled CKA pairs stages by relative progress across different layer counts. That is a descriptive comparison, not recovered variable correspondence. Sources are learning summary [S08], geometry [S09], and sequence splits [S10].



Three-seed learned-model accuracy with sequence-disjoint tests and a majority baseline.

12 Learned versus compiled geometry

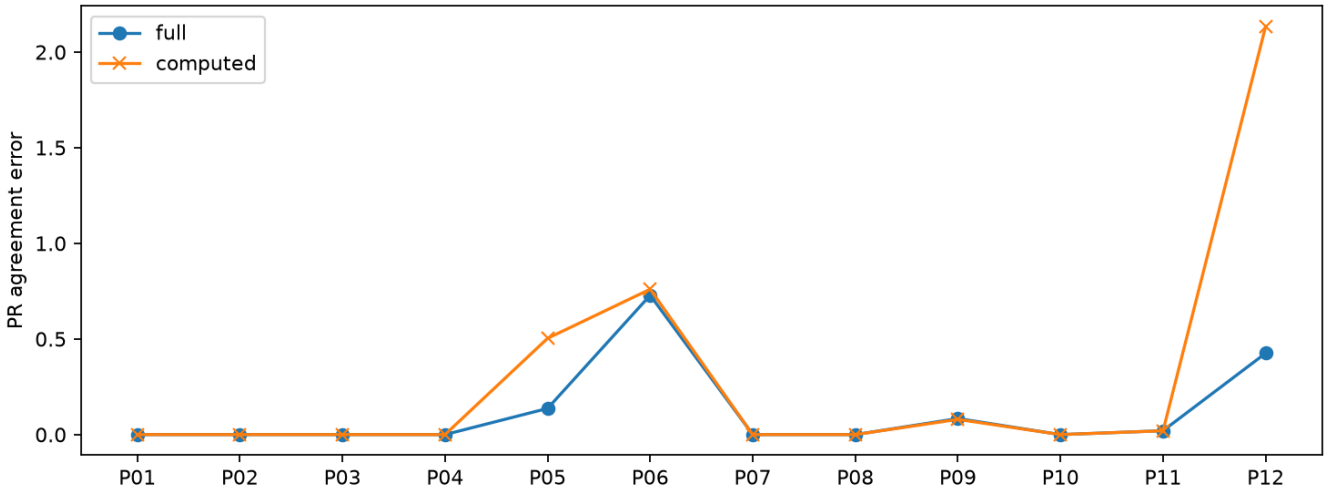
Final-stage representation comparison, averaged over seeds, follows. Different feature spaces and unknown learned circuits prevent interpreting PR or CKA agreement as algorithm recovery.

Program	Learned PR	Compiled PR	CKA
P01	3.56967	10.6236	0.715386
P02	1.01726	7.32466	0.802356
P03	1.29963	18.8703	0.362354
P04	4.2158	17.7164	0.45823
P05	1.13529	13.9161	0.352924
P06	1.54495	13.3776	0.497534
P07	4.20361	19.6008	0.396962
P08	4.57286	21.1789	0.355011
P09	1.28516	17.3664	0.198811
P10	6.15651	14.9264	0.400056
P11	2.03145	14.8206	0.152299
P12	1.56001	26.8434	0.338914

13 Original compiler-fidelity checks

Expected intermediate values are independently RASP-evaluated, then encoded with compiler-provided coordinates and write times. This is a fidelity benchmark. PR agreement is secondary because output cardinality, scaling, and covariance affect effective dimension.

Program	PR error	Lane agreement	CKA error	Defined pairs	Transitions	Definition mismatches
P01	0	1	N/A	0	2	0
P02	0	1	N/A	0	2	0
P03	0	1	N/A	0	2	0
P04	8.88178e-16	1	9.99201e-16	1	2	0
P05	0.503384	1	N/A	0	2	1
P06	0.759985	1	N/A	0	2	1
P07	2.82395e-07	1	4.75663e-08	2	6	0
P08	2.22315e-07	1	5.08566e-08	2	6	0
P09	0.0794928	1	0.00278887	5	6	0
P10	1.33612e-07	1	3.92533e-08	2	6	0
P11	0.019994	1	0.00038537	8	12	0
P12	2.13388	0.984127	0.0855042	4	6	0



Full and computed residual views on the same matched observations. Computed views retain compiler helper lanes.

14 Relation to InterpBench

Gupta and colleagues' InterpBench paper [S11] trains semi-synthetic transformers with known circuits using Strict Interchange Intervention Training and evaluates circuit-discovery methods. Its implementation [S12] supports ACDC, subnetwork probing, and edge attribution patching. RaspFormer evaluates representation measurements and transition detection, with additional supervised learned-model comparisons. Its learned models use ordinary output supervision and do not inherit known internal causal circuits. No InterpBench model or circuit-discovery benchmark was run here.

15 Evidence and limitations

Every study has a completed canonical manifest under `results/<study>/reference`. New disposable runs live under `results/<study>/runs/<UTC>-<id>`. Manifests record numerical policy, dependencies, compiler revision, source hashes, notebook source hashes, and Git state. A dirty Git state is recorded explicitly.

The new experiments do not prove universal computation recovery. Learned models lack internal ground truth. The compressed models retain attention and MLP hidden dimensions. Transition detection does not identify variable names. PCA-to-variable decoding uses supervised labels after geometry selection. No experiment establishes variable erasure or general lifetime recovery. Negative results and unavailable metrics remain in the data.

16 Verification and independent review

These are recorded checks from the completed implementation run, not fresh experiment executions performed to typeset this PDF. PDF layout and text were checked separately after generation.

Check	Recorded outcome
Original baseline	8 tests passed before implementation changes.
Full implementation suite	28 tests passed in 33.884 seconds. A focused seven-test research suite also passed after the scientific corrections.
Notebook execution	All 5 notebooks and all 37 code cells passed.
Learned-model replay	36 saved checkpoints reproduce their saved predictions and independent RASP targets on 677 held-out sequences.
Canonical evidence	7 completed study manifests; 12 research programs verified.
Frozen predictions	24 original / transformed prediction artifact hashes checked; 288 transformed detector result rows.
Patch stage selection	All 12 selected stages independently recomputed from calibration activations only.
Independent review	Model, scientific, and final audits passed after the accepted corrections.

Corrections made during review

- Added the always-positive write baseline. CKA on P11 scores below that baseline despite a favorable pooled result.

- Disclosed that decoder calibration folds share an unsupervised PCA fit. Separate test sequences remain held out.

- Moved patch-stage selection from unlabeled test activations to calibration activations, then reran the controls.

- Labeled transformed-recovery results as held-out-program macro means and retained task accuracy beside intended-schedule F1.

- Strengthened verification to recompute the selected-stage argmax and derive checkpoint coverage from the checked list and manifest.

The final audit was performed by gpt-6-astra. Earlier model and scientific reviewers used gpt-6-astra and gpt-6-sol. The review covers execution and evidence consistency; it does not turn the limited experiments into a general scientific guarantee.

Recorded verification, review notes, and the append-only decision trail are preserved in verification evidence [S13], review notes [S14], and the decision trail [S15].

Reproduction commands

```
uv python install 3.12
uv venv --python 3.12 .venv
uv pip install --python .venv/bin/python -r requirements.txt
.venv/bin/python -m raspformer.study
.venv/bin/python -m raspformer.report
mkdir -p .context/verification
.venv/bin/python -m unittest discover -v > .context/verification/tests.log 2>&1
.venv/bin/python scripts/check_notebooks.py
.venv/bin/python scripts/replay_learning.py
.venv/bin/python scripts/verify_results.py \
  --test-log .context/verification/tests.log \
  --notebook-summary .context/notebook-checks/summary.json
```

To rerun a subset, select `--sections baseline`, `--sections validation`, or `--sections learning`. The learning defaults are seeds 42, 43, and 44 with 600 optimization steps. Reference promotion occurs only after successful numerical completion.

17 Evidence index

Paths below are relative to the repository root. They identify the raw evidence behind this standalone document. The PDF contains the results and interpretation; the repository contains the full per-observation tables, activation arrays, model weights, and scripts.

ID	Evidence	Repository path or primary source
S01	matched stages	results/comparison/reference/matched/stages.csv
S02	floor sensitivity	results/comparison/reference/matched/variance-sensitivity.csv
S03	balanced sensitivity	results/comparison/reference/balanced/variance-sensitivity.csv
S04	the validation reference	results/validation/reference
S05	every scalar error	results/validation/reference/p12-sorting/per-observation-errors.csv
S06	attention sweep	results/validation/reference/p12-sorting/attention-sharpening.csv
S07	failing probe identities	results/validation/reference/p12-sorting/failing-probe-sets.json
S08	learning summary	results/learning/reference/summary.csv
S09	geometry	results/learning/reference/geometry.csv
S10	sequence splits	results/learning/reference/sequence-splits.csv
S11	InterpBench paper	https://papers.nips.cc/paper_files/paper/2024/hash/a8f7d43ae092d9a5295775eb17f3f4f7-Abstract-Datasets_and_Benchmarks_Track.html
S12	implementation	https://github.com/FlyingPumba/InterpBench
S13	verification evidence	results/validation/reference/verification.json
S14	review notes	results/report/review.md
S15	decision trail	results/report/decisions.tsv
S16	Repository overview	README.md
S17	Program outputs and allocations	results/programs/reference/manifest.json
S18	Manual circuit traces	results/trace/reference/traces.json
S19	Balanced geometry metrics	results/geometry/reference/layer_metrics.csv
S20	Original lane interventions	results/alignment/reference/interventions.csv
S21	Compiled diagnostics and transformed-model summaries	results/validation/reference/summary.json
S22	Depth and branch-count families	results/validation/reference/complexity/factorial-families.csv
S23	Checkpoint replay evidence	results/learning/reference/replay.json

18 Decision trail

The following entries summarize the implementation decisions in chronological order. Two initially open entries were resolved by later review and completion entries. Full timestamps, reasons, and source pointers remain in results/report/decisions.tsv.

UTC, 15 Oct	Phase	Decision	State
07:10:37	frame	Use bounded CPU experiments with explicit success and failure results	open
07:30:32	baseline	Original eight tests passed before edits	VERIFIED
07:30:32	design	Choose one executable transformer for all model variants	VERIFIED
07:30:32	numerics	Apply total sample variance floor of 1e-12 and retain raw metrics	VERIFIED
07:30:32	learning	Train all 12 programs with three random seeds and 600 Adam steps	VERIFIED
07:30:33	diagnosis	Reject identical-failing-row assumption for P12	VERIFIED
07:30:33	review	Add always-positive write baseline after independent scientific review	open
07:39:19	verification	All 28 tests and all five notebooks passed	VERIFIED
07:39:19	replay	Replay all 36 learned checkpoints against saved predictions and independent RASP targets	VERIFIED
07:39:19	cleanup	Remove 27 outdated report and archive files and replace canonical references	VERIFIED
07:39:19	review	Report always-positive baseline and shared calibration PCA explicitly	VERIFIED
07:41:01	validation	Rerun blind recovery on executing rotated and compressed variants	VERIFIED
07:41:01	complete	Resolve the initial open frame with completed experiments and verification	VERIFIED
07:47:07	correction	Move PCA patch stage selection from unlabeled test activations to calibration activations	VERIFIED
07:47:08	verification	Recompute frozen calibration selections and verify all evidence after the correction	VERIFIED
07:48:17	review	Final independent audit passed and verifier maintenance suggestions were applied	VERIFIED

The remaining scientific boundaries are explicit. Original alignment uses compiler knowledge. Blind detection recovers write events, not complete algorithms. Reduced models may implement changed computations. Learned internal circuits are unknown. No general erasure, lifetime, length-generalization, or natural-language result is established.